

Navigating Factor Crowding: A Research Roadmap

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Abstract

This paper explores the evolving dynamics of factor investing in quantitative equity portfolio management, particularly addressing the challenges posed by increased factor crowding and proposing strategies for adaptation. It draws on John Maynard Keynes's Beauty Contest analogy to conceptualize the strategic navigation of alpha research, emphasizing the necessity to anticipate the collective bias of market participants. The "quant meltdown" of 2007 and "quant winter" of 2018-2020 underscored the pitfalls of traditional alpha research that failed to account for the influence of collective investment behaviors, leading to frequent factor crashes and a non-stationary market environment. We propose a shift of research focus towards second-degree alpha research, which considers the actions of other investment managers, thus aiming to exploit patterns emerging from collective trading behaviors. This paper also discusses the ramifications of factor crowding, likened to traffic dynamics, where advanced knowledge of market movements can be used to exploit arbitrage opportunities. Ultimately, this paper seeks to redefine the approach to factor investing by integrating insights into crowding dynamics, risk management, and the differentiation between risk and uncertainty to navigate the complexities and challenges of today's markets.

Factor investing is the cornerstone of quantitative equity portfolio management. Quantitative investment managers strive to outperform their benchmarks through meticulous factor research, aiming for higher returns with reduced risk. As factor investing gained traction, the industry began experiencing more frequent factor return crashes. The period between 2018 and 2020 saw such significant and sustained underperformances that it was dubbed the "quant winter" by practitioners. Ten years prior, quants also experienced a meltdown in August 2007. Jermey Stein (2009) attributed the 2007 meltdown to crowded trade and overleverage. He called it a coordination problem among sophisticated investors and suspected the crowding phenomenon would persist.

In this paper, we propose that the dynamics of factor investing are evolving due to increased factor crowding. We explore how quantitative managers should adapt their factor research in response to this shifting terrain in order to seek strategic advantages. We question how to best position factor research and what might be expected from factor investing over the next decade.

This paper is structured as follows: it begins with an analysis of the changing environment using Keynes's Beauty Contest as a conceptual framework, reviews the historical focus of factor research, suggests necessary adjustments, and provides analogies. Data-driven research outcomes will be published in future companion papers.

Keynesian's Beauty Contest (Keynes, 1936)

We use Keynesian's Beauty Contest to guide our discussion of the evolution of alpha research, and here is a brief overview.

John Maynard Keynes illustrated the behavior of rational market participants with an analogy of a hypothetical newspaper contest. In this contest, participants are asked to select six faces they find most attractive from a set of one hundred photographs. The winners are those whose choices align most closely with the most popular selections.

While a simple approach might involve choosing the faces one personally finds most attractive, a more strategic entrant would consider the general public's perceptions of attractiveness. By estimating what the majority might prefer, based on an understanding of societal standards and trends, the entrant enhances her chances of selecting the most popular faces and, consequently, winning the prize.

"It is not a case of choosing those [faces] that, to the best of one's judgment, are really the prettiest, nor even those that average opinion genuinely thinks the prettiest. We have reached the third degree where we devote our intelligences to anticipating what average opinion expects the average opinion to be." (Keynes, 1936)

Traditional Alpha Research

Traditional alpha research focuses on uncovering asset pricing anomalies by analyzing historical data to identify patterns that previously yielded high information ratios (IR). This approach, grounded in first-degree rationality, aims to exploit these patterns to achieve excess returns with minimal risk. Quantitative researchers, in essence, seek the most attractive investment opportunities, mirroring John Maynard Keynes's concept of selecting "the prettiest" faces based solely on individual judgment.

To gain an edge, factor researchers enhance their strategies by addressing three pivotal questions: where to find more consistent returns cross-sectionally; when to capitalize on these opportunities over time; and how much risk and liquidity to incorporate at each portfolio rebalance. Essentially, one tries to develop a better "alpha trap".

Although traditional alpha research is commonly accepted by practitioners, its Achilles heel is that it does not consider the actions of other investment managers. This paradigm of thinking can be described as "my data and I in my office." When quants perform historical simulations, it might feel as if they are the only ones arbitraging a repeating pattern in the data. This myopic first-person view, in our opinion,

leads to crowding, frequent crashes, and non-stationarity, as we will illustrate shortly.

Alpha Research of the Second Degree

When alpha research is advanced to a higher order of rationality, it aims to capitalize on patterns that arise from the collective trading behaviors of other investment managers who operate under first-order rationality. This second-order rationality seeks to exploit opportunities that materialize from the aggregated actions of these managers, aligning with Keynes' idea of selecting "those that average opinion genuinely thinks the prettiest," or in this context, popular investment opportunities.

To differentiate from the behavioral angle, first-order research primarily focuses on exploiting profit opportunities that arise from the suboptimal behaviors of various market participants like retail investors, corporate managers, sell-side analysts, and government agencies. These behaviors are well-documented in the asset pricing anomaly literature and form the basis of traditional quantitative strategies. As more quantitative managers draw from this same body of literature, use similar datasets, employ standard back-testing methodologies, and rebalance their portfolios with comparable frequency and turnover, predictable patterns begin to surface. Security brokerage firms are particularly adept at recognizing and acting on these patterns through activities like trading prediction and supplying liquidity, often on an intraday and high-frequency basis. In contrast, we believe factor crowding has broader implications than just the high-frequency basis, as we elaborate in the following sections.

The Anatomy of Factor Crowding

Given the preceding background information, let's formally analyze the origin of factor crowding, the set of new challenges, and associated research priorities. It starts with first-order alpha research, the paradigm of "my data and I in my office." Quant researchers typically conduct research in isolation; at times, it may feel that they are the only ones arbitraging historical patterns in data. However, if one can broaden her

field of vision, elevate her vantage point to a high level, and look back down at the vicinity of her office, the illuminating view will show that many quant shops are doing the same thing, albeit in different isolated locations. Investment consultants are aware of this. During their annual visits to various managers, they frequently encounter firms presenting proprietary research that, ironically, overlaps with what other visited firms are presenting. Naturally, the term, common proprietary research, seems paradoxical, originating from the narrow perspective of "my data and I in my office."

Had quants been doing similar research/trading independently, the consequences of crowding would not be as severe. Unfortunately, quant research/trading exhibit high levels of correlation among firms. First, quants use the same data sources from the same set of data vendors. As data vendors promote their "new" dataset to boost sales growth each year, firms often find themselves undertaking similar research projects each year. Second, the statistical methods employed for historical simulations are based on the same mathematical principles. While individual parametrization may vary, the underlying principles remain the same. A phenomenon discovered by one firm is likely to be found by another should programming exercises be conducted correctly. Third, quant firms balance their portfolio with similar turnover and frequencies. Since transaction cost and risk models are nearly identical across third-party vendors, given that they are based on the same set of data, an extensive historical simulation will yield similar optimal turnover and trading frequencies. Fourth is synchronized trading. Fundamental data are released through earnings announcements, typically four times a year. Most companies release their earnings within the same time frame. For example, eighty-five percent of S&P 500 companies release their earnings within a three-week window. It takes a few hours to a few days for a company's new fundamental data to filter through the data feed. Based on these newly released data, quants tend to buy companies with good earnings and sell those with bad ones. Hence, both the trade direction and timing are synchronized. Such implicit synchronization mechanism exists in other

datasets, such as earnings estimates, geographical revenues, and supply chains, to name a few.

Moreover, the phenomenon of factor crowding is directly attributed to the extreme liquidity pressure within a short period of time. It bears some similarity to the phenomenon of gravitational singularity, a.k.a. black hole, which as the following excerpt from Wikipedia describes, "... is a condition in which gravity is predicted to be so intense that spacetime itself would break down catastrophically. As such, a singularity is by definition no longer part of the regular spacetime and cannot be determined by 'where' or 'when'". Similarly, when liquidity pressure becomes extreme, both the fundamental and quantitative understanding of stock price behavior breaks down. This gives rise to a new set of challenges for factor investing: 1) diminishing factor returns; 2) more frequent factor return crashes; and 3) non-stationarity.

We advocate that the second degree of research should address the emerging challenges while also targeting new opportunities. The research outcome aims to arbitrage the collective behaviors of managers who solely focus on traditional alpha research of the first degree. In the following sections, we will discuss each of these focus areas in detail.

Factor Crowding Arbitrage

Traffic jams provide a useful metaphor for understanding the dynamics of factor crowding

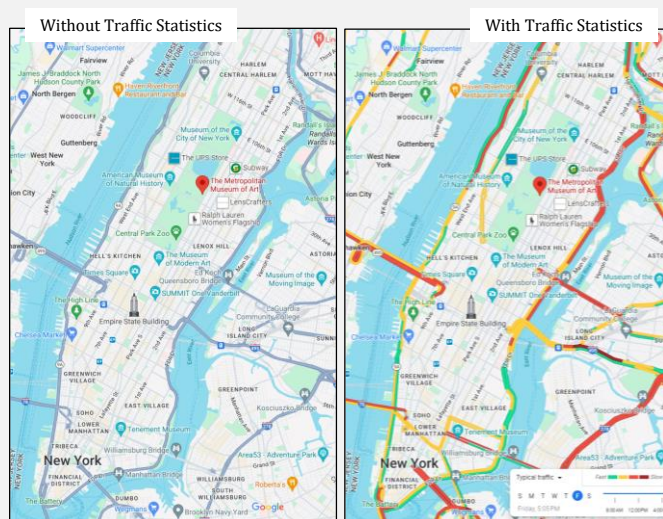
and the opportunities for arbitrage. To illustrate this concept, let's consider a hypothetical scenario involving Google Maps. Starting tomorrow, Google introduces a two-tiered service model for its navigation app. The basic, free version provides comprehensive services but omits real-time traffic updates. Conversely, the premium, paid version includes all features, notably real-time traffic information.

Imagine a rational New York City commuter using the free version of Google Maps, as depicted on the left panel of Exhibit 1. She decides to leave work an hour early on a Friday afternoon, opting for the shortest route to aim at a swift exit from the city. However, since many other commuters with the same version of the app make similar decisions, they all converge on the same route at the same time. This results in significant traffic jams and delays, exactly what they hoped to avoid.

In contrast, a commuter with access to the paid version of Google Maps can see the current traffic conditions, as shown on the right panel of Exhibit 1. This allows her to adapt her route in real-time, steering clear of congested areas. Her goal is to find the fastest route, not necessarily the shortest. This ability to dynamically adapt her path based on live data helps her navigate out of the city more efficiently, demonstrating the clear advantage of having access to the congestion information.

This scenario illustrates how access to different levels of information can create

Exhibit 1 | Hypothetical Scenario of Google Maps App



opportunities for arbitrage, where individuals with superior information can optimize their decisions and outcomes more effectively than those with restricted access.

Crowding Statistics

The counterfactual scenario underscores the critical role of real-time information for commuters aiming to exit the city swiftly. Similarly, in the financial world, access to real-time crowding statistics provides a competitive edge in arbitraging crowded trades. Despite its intuitive appeal, factor crowding is difficult to quantify formally. Different methodologies offer insights into this phenomenon; some researchers suggest using correlated holdings among institutional investors as an indicator of crowding (Lazo et al., 2024), while others look at trading patterns and mutual fund flows to assess crowded trading (Greenwood et al., 2011).

These approaches generally rely on a technical analysis premise that stock trading drives price changes. However, contemporary finance theories posit that stock prices are fundamentally tied to intrinsic values, fluctuating randomly around these values. As new information—like earnings surprises—emerges, intrinsic values assimilate the information and adjust to new values. Thus, trading contributes to price discovery but does not set the price itself. Research focusing on measurements of crowding and its new discoveries can be invaluable for future factor investing.

Understanding the crowding dynamics allows investors to exploit the behavior of other market participants. For instance, in the context of price momentum (PM), stocks can be categorized into crowded and non-crowded groups. Crowded PM stocks, often well-recognized by investors and potentially overvalued due to their popularity, contrast sharply with non-crowded PM stocks. These lesser-known stocks might be in the early stages of their momentum lifecycle, presenting a prime investment opportunity. A strategic investor may opt to buy these under-the-radar stocks while selling off the overexposed ones, effectively capitalizing on the differential market dynamics between the two groups.

Factor Crash Insurance

In this section, we delve into the recurring phenomenon of factor crashes, with a specific focus on risk-on crashes. We explore their triggers, measure the risk factors involved, and discuss strategies for mitigation akin to insurance policies.

Risk-on Crash: Understanding Triggers and Dynamics

A notable recurring theme in factor crashes is related to the cyclical changes in investors' risk appetite, oscillating between risk-on and risk-off moods. This cycle is largely driven by investors' expectations about future economic growth and stock market trends. In a risk-on phase, investors, buoyed by optimism about the economy, tend to

Exhibit 2 | The Risk-On, Risk-Off Cycle



purchase stocks of cyclical companies with high stock price volatility, shedding their holdings in more stable, low-volatility stocks as illustrated on the left teardrop of Exhibit 2. Conversely, a risk-off mood sets in when concerns about economic downturns prevail, prompting investors to divest from economically sensitive, high-volatility stocks and to seek refuge in low-volatility stocks, as depicted on the right teardrop of Exhibit 2.

During the cyclical shifts between risk-on and risk-off phases, two critical inflection points emerge, each marked by significant market activity that often resembles a crowded, stampede-like rush in trades. These transitions can drastically affect factor returns, particularly influencing quantitative investment strategies.

Quantitative models typically adopt a strategy of longing low-volatility stocks while shorting high-volatility ones. The rationale behind this approach is that low-volatility stocks usually provide more stable and consistent returns over time compared to their high-volatility counterparts. This stability makes them appealing to the procedure of historical simulation.

However, this implicit bet against volatility becomes particularly vulnerable at the risk-on inflection points. These are the moments when the market sentiment shifts towards a more aggressive stance, with investors anticipating economic growth and a bullish market. As optimism grows, there's a significant pivot in trading behavior, where investors start buying high-volatility stocks, driven by the potential for higher returns. This shift is depicted at the bottom of the Yin-Yang cycle in Exhibit 2.

Measuring the Risk: The Volatility Factor

Stock volatility serves as a quantifiable measure of these mood swings, measuring the historical return volatility of each stock. It captures investors' shifting preferences between high- and low-volatility stocks and effectively characterizes the sentiment-driven market dynamics. In this paper, the volatility risk factor is used to gauge the propensity of factor crashes, around the inflection points.

Insurance in the Factor Space: Premiums and Coverage

Quantitative models can employ a hedging strategy to mitigate the risk associated with a

volatility-driven factor crash at the risk-on inflection point. This strategy involves buying a portion of the volatility factor into the alpha model—specifically, long positions in high-volatility stocks and short positions in low-volatility stocks. This strategy aims to counterbalance the model's inherent tendency to bet against high-volatility stocks.

Drawing parallels to traditional insurance, in the factor investment space, the "insurance premium" resembles the expected return of a risk factor. When an investor incorporates a risk factor into their portfolio to mitigate against potential crashes, the expected return of that factor represents the cost (or benefit, if positive) of the coverage. The "insurance coverage," on the other hand, is represented by the time-series return correlation between the insured factor and the risk factor.

To optimize Factor Insurance, an investor must weigh the cost (premium) against the degree of protection (coverage) provided by different insurance policies. The ideal scenario is to find an insured factor that not only offers a high positive correlation with the risk factor but also delivers a positive expected return—essentially the investor is paid to insure against the risk. Developing an optimization framework that balances expected returns with the correlation of an insurance policy could significantly mitigate recurring factor crashes. Unlike traditional mean-variance optimization, which focuses on balancing returns with variance, this approach would tailor the trade-offs specifically to factor crash risks, aiming to identify the most effective insurance policy for any given risk factor. By employing these strategies, quantitative managers can better navigate the inherent risks of cyclical market sentiments, securing their investments against potential losses while optimizing returns.

Modeling Non-Stationarity Explicitly

The mean-variance efficient portfolio, as pioneered by Harry Markowitz's Modern Portfolio Theory (MPT), has long been a cornerstone of financial forecasting. However, the effectiveness of this approach hinges on the assumption that financial markets are

stationary—that is, parameters estimated in one period are expected to persist into another. This assumption presumes that history not only informs but also reliably predicts the future without significant deviation. When the reality of non-stationarity prevails—meaning market behaviors and structural relationships can and do change—the traditional mean-variance strategy often falls short, resulting in suboptimal performance ex-post.

Analogy of Non-Stationarity

Non-stationarity in financial markets can be likened to walking in New York City on a windy, rainy day. As the wind interacts with skyscrapers, it naturally follows the path of least resistance. As downward motion accelerates down buildings, particularly around corners, its effects become most pronounced. With dozens of buildings nearby, each with its own path of least resistance, chaotic and unpredictable nature of wind direction is magnified. Imagine walking along New York City streets on such a windy, rainy day, holding an umbrella in hand in an attempt to stay dry. As strong winds blow in one direction, one positions the umbrella against the wind to shield herself from the rain. Just as one thinks, “I got this”, the wind suddenly shifts direction unexpectedly and blows the umbrella inside out. Similarly, quantitative strategies can suffer severe underperformance due to sudden, unforeseen shifts in market conditions—deviations that traditional risk models often fail to anticipate. For example, a 3% tracking error portfolio can lose 6% in a single month! Such an event should happen once in a million years, but we see them occasionally in our lifetime.

Crowding and Non-Stationarity

Leverage plays a critical role in the dynamics between crowding and non-stationarity within financial markets, particularly evident in the analysis of events like the 2007 quant meltdown (Stein, 2009; Khandani 2011). This relationship can be illustrated through a hypothetical scenario involving three hedge funds—A, B, and C—which engage in leveraged investments to amplify their potential returns, focusing on the same alpha factor and maintaining similar positions.

Scenario Breakdown

- Initial Shock to Fund A: Suppose Fund A experiences an idiosyncratic shock—a large

client unexpectedly withdraws their investment. To meet the sudden liquidity demand, Fund A is forced to sell its positions, both long and short. This massive, unplanned sell-off negatively impacted the factor they were arbitraging, causing factor returns to plummet.

- Margin Call on Fund B: Fund B, which operates with higher leverage than Fund C, is immediately affected by the downturn in factor performance initiated by Fund A's liquidations. The decline in the value of its positions triggers a margin call. To satisfy the margin requirements, Fund B must quickly liquidate assets, exacerbating the downward pressure on factor returns through a fire-sale effect.
- Cascading Effect on Fund C: As Fund B's liquidations continue to deteriorate the factor's performance, Fund C finds itself in a similar predicament. It, too, faces a margin call and is compelled to sell off its holdings, further driving down factor returns and perpetuating the fire-sale dynamic.

The interconnectedness due to similar investment strategies (crowding) combined with high leverage means that an issue in one fund can quickly propagate through the financial system, leading to significant market disruptions. This is a clear demonstration of non-stationarity, where past statistical relationships and behaviors (like the stability of a particular factor's returns) do not hold under current or future conditions due to external shocks.

Risk vs. Uncertainty

To address non-stationarity effectively, it is crucial to distinguish between risk and uncertainty—a concept famously discussed by economists Frank Knight and John Maynard Keynes. Knight differentiated risk (which can be measured and quantified) from uncertainty (which cannot be measured and involves unforeseeable changes). Keynes further emphasized that true uncertainty pertains to situations where no calculable probability can be applied—essentially, the unknown unknowns.

Frank Knight (1921) once said,

“Uncertainty must be taken in a sense radically distinct from the familiar notion of Risk,

from which it has never been properly separated... The essential fact is that 'risk' means in some cases a quantity susceptible of measurement, while at other times it is something distinctly not of this character; and there are far-reaching and crucial differences in the bearings of the phenomena depending on which of the two is really present and operating.”

Risk, in the mean-variance framework, is quantified by the covariance matrix, comprising measurable quantities determined a priori. Knight is cognizant that a different driver, an unquantifiable phenomenon, is at work in the real world—the so-called uncertainty.

John Maynard Keynes (1936) went further and said,

“By ‘uncertain’ knowledge, let me explain, I do not mean merely to distinguish what is known for certain from what is only probable. The game of roulette is not subject, in this sense, to uncertainty...The sense in which I am using the term is that in which the prospect of a European war is uncertain, or the price of copper and the rate of interest twenty years hence...About these matters there is no scientific basis on which to form any calculable probability whatever. We simply do not know.”

By distinguishing the idea of uncertainty from risk, in “we simply do not know”, Keynes confesses the limits of human knowledge, as opposed to the idealistic view that the market is completely intelligible.

Incidentally, Donald Rumsfeld seemed to sum it up most succinctly, in a way a six-year-old can understand,

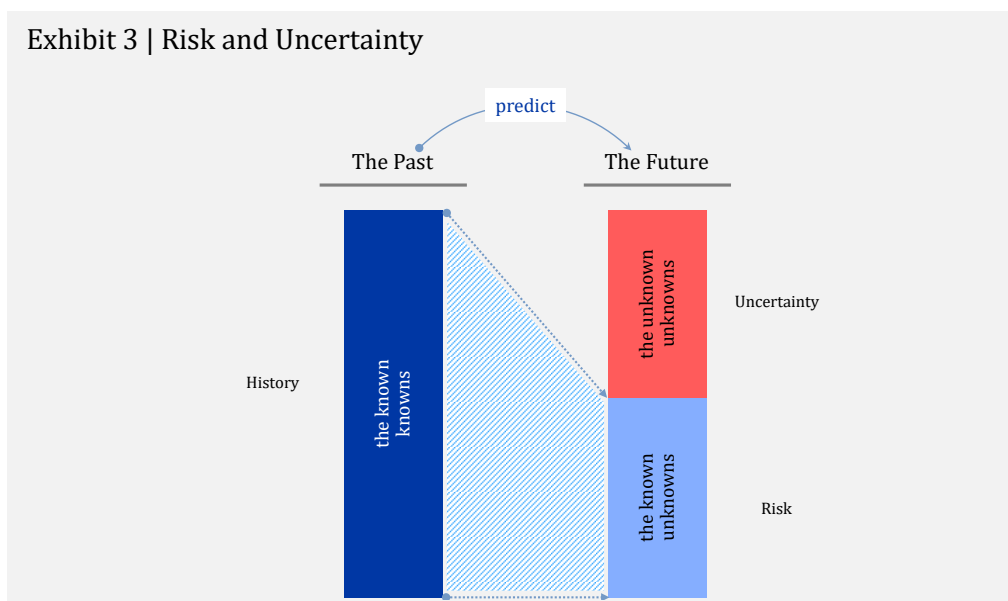
“...there are known knowns; there are things we know we know. We also know there are known unknowns; that is to say, we know there are some things we do not know. But there are also unknown unknowns—the ones we don’t know we don’t know.”

This distinction is vital because traditional financial models, like those using the mean-variance framework, are designed to manage risk through quantifiable metrics such as covariance matrices. They are ill-equipped to handle uncertainty, which encompasses unquantifiable phenomena and events beyond the scope of statistical predictability.

Towards a New Forecasting Framework

To overcome the limitations of traditional models, quantitative research must develop frameworks that accommodate both risk (the known unknowns) and uncertainty (the unknown unknowns). Such models would not only rely on historical data but also integrate measures to adjust for the potential non-stationarity of market conditions. By acknowledging and planning for uncertainty, researchers can create more robust forecasting models that are better suited to predict and perform in real-world scenarios, as illustrated in Exhibit 3.

This approach involves tuning models to the specific degree of non-stationarity observed in



different markets, and balancing the influence of both risk and uncertainty. Such a refined method aims to improve out-of-sample performance by adapting to the inherent unpredictability and variability of financial markets.

In summary, while traditional mean-variance efficient portfolios offer a foundational approach to investment strategy, their reliance on the

stationary assumption can lead to significant drawbacks in a non-stationary world.

Recognizing and differentiating the impacts of risk and uncertainty allows for the development of more adaptive and resilient quantitative strategies that better reflect the complex realities of today's financial landscapes.

Conclusion

In conclusion, this paper underscores the critical need for a paradigm shift in quantitative equity portfolio management, particularly in the realm of factor investing. As evidenced by the challenges highlighted during the "quant winter" period, traditional alpha research strategies, which focus solely on historical data to predict future returns, are increasingly inadequate in today's crowded markets. By adopting Keynes's Beauty Contest framework, we illustrate the importance of evolving from a first-degree to a second-degree alpha research approach, where understanding and anticipating the collective actions of other market participants becomes paramount.

We discuss the pervasive issue of factor crowding and its implications on market stability and the efficacy of quantitative strategies. The analogy of traffic dynamics, particularly through the use of a counterfactual example of Google Maps, serves to elucidate how real-time data and a deeper understanding of collective behaviors can offer significant arbitrage opportunities, thereby enhancing portfolio performance and resilience against crowded market conditions.

Moreover, this paper highlights the necessity of addressing non-stationarity and differentiating between risk and uncertainty in investment models. As factor investing continues to evolve, the ability to adapt to and anticipate changes in market behavior—not merely based on historical patterns but also through a higher level of reasoning—will be crucial for sustaining competitive advantages.

In essence, the future of factor investing must embrace a comprehensive, forward-thinking approach that not only seeks to understand the current landscape but also prepares for the uncertainties of tomorrow. By integrating these advanced research methodologies and strategic insights, quantitative fund managers can better shield their investments from systemic risks and exploit new opportunities for growth and stability in an increasingly complex market environment.

Disclosures

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This paper contains certain forward-looking statements which can be identified by the use of forward-looking terminology such as "may," "will," "should," "expect," "anticipate," "project," "estimate," "intend," "continue," or "believe," or the negatives thereof or other variations thereon or comparable terminology. Such statements are based on current expectations, estimates and projections about the industry and markets in which the authors invest, as well as the authors' beliefs and assumptions. These statements involve certain risks, uncertainties and assumptions which are difficult to predict. Consequently, such statements cannot be guarantees of future performance and actual outcomes and returns may differ materially from statements, goals and objectives set forth herein.

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